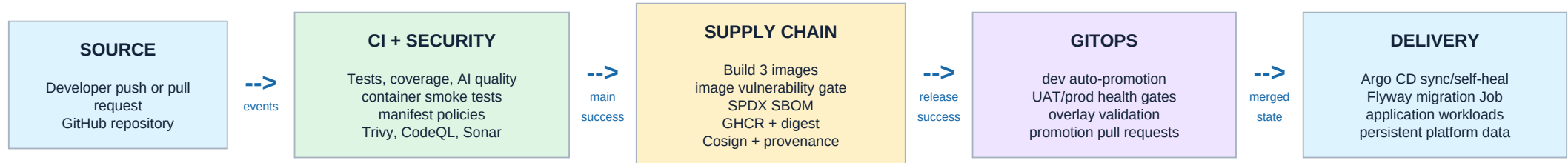
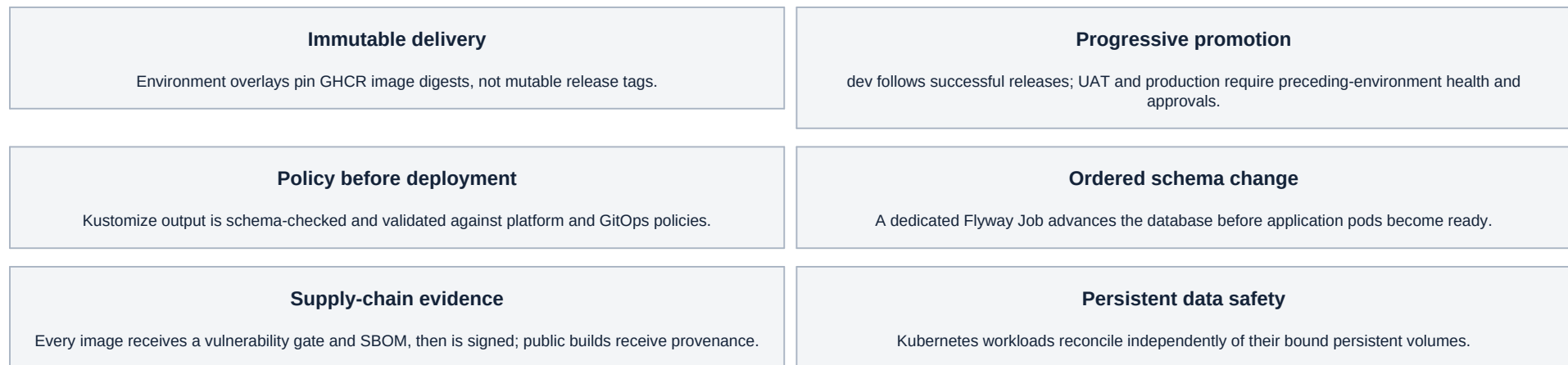


CI/CD pipeline architecture

GitHub Actions validates and secures every change, GHCR distributes signed application images, Git pull requests promote immutable digests, and Argo CD reconciles Kubernetes environments.



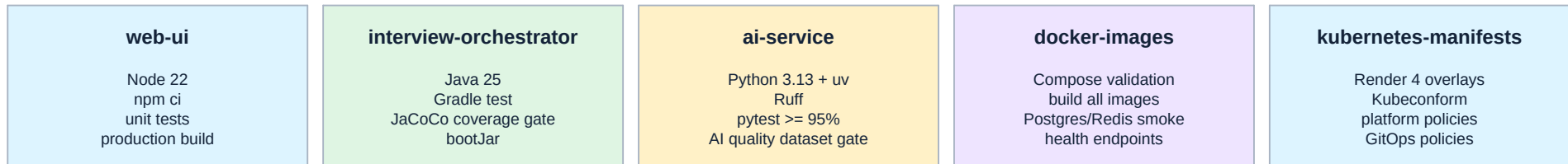
Core design guarantees



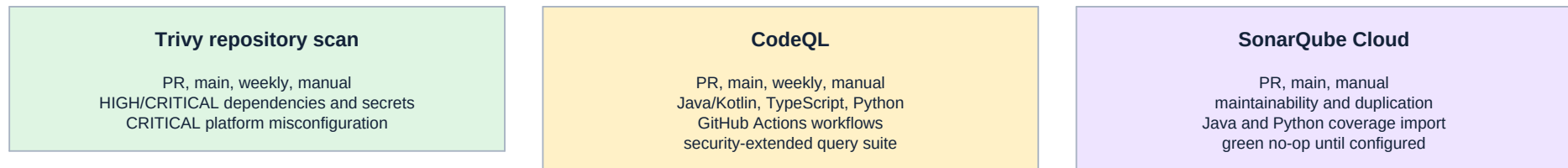
1. Continuous integration and security

Pull requests exercise the same central quality gates used on main. Independent security workflows add repository, source-code, and maintainability coverage.

platform-ci



Independent security and quality analysis

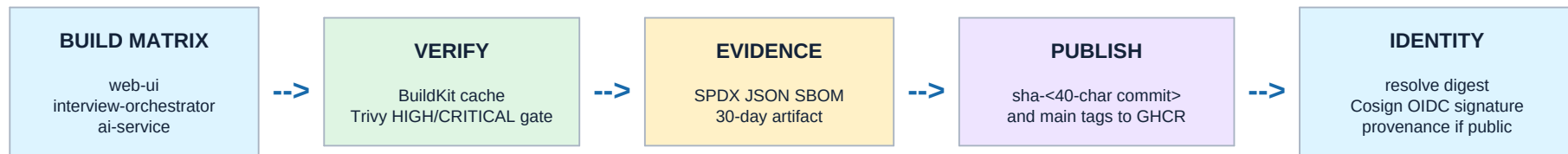


Trigger rule: only a successful platform-ci run associated with main automatically starts release-images. Digest-only GitOps commits are detected and skipped, preventing a recursive release loop.

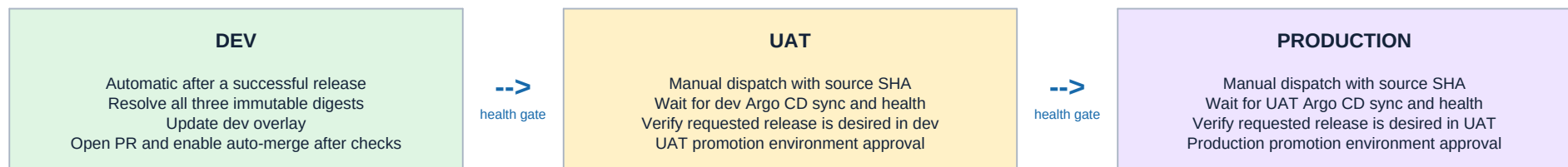
2. Release and progressive promotion

The release pipeline creates auditable artifacts once, then promotes the exact same image digests between environments.

Software supply chain



Environment promotion

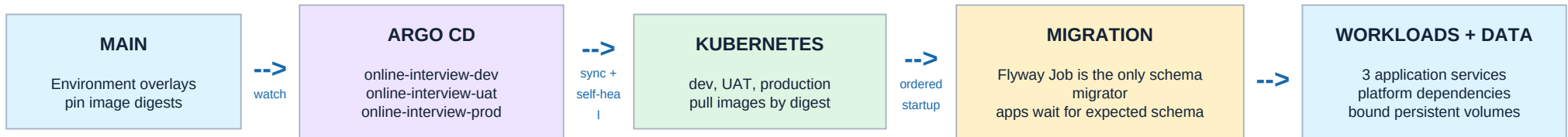


Every target overlay is rendered, checked with Kubeconform, and validated by GitOps policy before its promotion pull request is opened. Merging the PR changes desired state on main; it does not directly mutate a cluster.

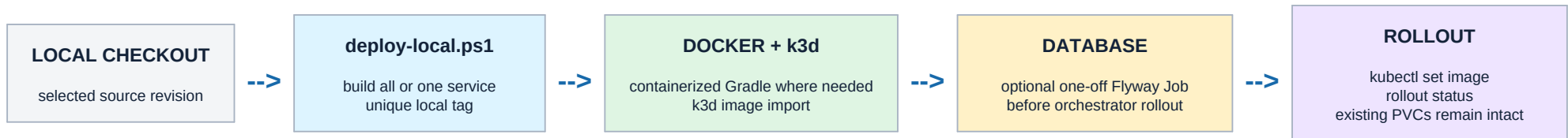
3. Argo CD delivery and local k3d path

Shared environments are declarative and GitOps-managed. Local development has a separate imperative build/import/rollout loop for rapid iteration.

Shared-environment delivery



Local k3d development



Repository sources of truth

Concern	Repository source of truth
CI gates	<code>.github/workflows/ci.yml</code>
Security	<code>.github/workflows/security.yml</code> and <code>codeql.yml</code>
Code quality	<code>.github/workflows/sonarcloud.yml</code>
Image release	<code>.github/workflows/release-images.yml</code>
Promotion	<code>.github/workflows/promote-gitops.yml</code>
Desired state	<code>platform/kubernetes/overlays/{dev,uat,prod}</code>
Argo CD	<code>platform/gitops/argocd</code>
Local deployment	<code>scripts/deploy-local.ps1</code>